JEEVAN BAABU MURUGAN

Computer Science Engineering — AI & Machine Learning

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate specializing in Artificial Intelligence and Machine Learning, with a strong foundation in software development, web technologies, machine learning, cloud computing, and IoT systems. Proficient in Python, JavaScript, TypeScript, React.js, and Next.js, with hands-on experience building scalable web applications, AI-powered tools, and edge-cloud hybrid systems end to end. A fast learner comfortable picking up new languages, frameworks, and tools as a role requires, and eager to contribute to full-stack, AI/ML, or cloud engineering roles.

EDUCATION

B.Tech, Computer Science Engineering — Artificial Intelligence & Machine Learning *Sep 2022 – May 2026*

SRM Institute of Science and Technology, Kattankulathur, Chennai

GPA: 8.29 / 10 (≈ 3.32 / 4.0) | First Class with Distinction

PROJECTS

EdgeSign — IoT-Enabled Edge–Cloud Hybrid Sign Language Translation

https://github.com/therealgone/EdgeSign-An-IoT-Enabled-Edge-Cloud-Hybrid-System-for-Real-Time

- Built a real-time sign language recognition system combining edge computing on a Raspberry Pi with cloud-assisted processing on a laptop for lower-latency inference at the edge.
- Captured hand gestures via camera and extracted landmarks using MediaPipe; trained a classifier to recognize predefined signs and translate them into words.
- Second-generation rebuild of an earlier gesture-recognition project (39-gesture prototype); redesigned the architecture around an edge-cloud hybrid pipeline for improved responsiveness and scalability.
- Tech Stack: Python, Raspberry Pi, MediaPipe, OpenCV, TensorFlow/Keras, Flask

CodeDock — AI-Assisted Multi-Language Code Runner

https://github.com/therealgone/CodeDock

- Designed and built an end-to-end platform where users generate code via AI or paste their own, select a language, and run it inside an isolated sandbox.
- Implemented Docker-based sandboxing with strict resource limits and no network access, running non-web languages in per-job containers while HTML/CSS/JS render live in-browser.
- Built an auto-fix loop: failed runs are sent back to the LLM with the error output, which retries generation up to a fixed attempt limit before returning final results.
- Tech Stack: Next.js, Express/FastAPI, Docker, OpenAI API, TypeScript

Acadia OS — Academic Portal Interface for SRM Academia

https://github.com/therealgone/Acadia-OS

- Developed a fast, mobile-first interface for SRM's academic system, consolidating timetable, attendance, marks, courses, calendar, and profile data into one dashboard.
- Built predictive attendance insights that project future attendance and show actionable margins for students.
- Implemented offline-friendly client-side caching with TanStack Query and a cookie-based authentication flow with middleware route protection.
- Tech Stack: Next.js, TanStack Query, Vercel Edge Middleware, Tailwind CSS

Medical Insurance Report Analysis System

https://github.com/therealgone/Medical-Insurance-Report-

- Built a Streamlit-based system that uses OCR and AI-driven text analysis to summarize medical reports into plain language for insurance claim processing.
- Integrated Google Gemini to interpret extracted text and flag key medical conditions and risk factors, generating structured, insurance-ready PDF reports.

TECHNICAL SKILLS

- Languages: Python, Java, C++, JavaScript, TypeScript
- Web & Mobile: React.js, Next.js, Node.js, Flask, HTML/CSS, Tailwind CSS, React Native
- AI/ML & Data: TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, MediaPipe, Pandas, NumPy
- Cloud, DevOps & Tools: Docker, Vercel, Netlify, Git/GitHub, REST APIs, MongoDB, MySQL, SQL
- Computer Vision: OpenCV, InnerEye-DeepLearning, MONAI, MedCLIP

CERTIFICATIONS & TRAINING

- JPMorgan Chase & Co. — Software Engineering Job Simulation (Forage) — Aug 2023
- Google for Developers — Android Developer Virtual Internship, via ACTE — Apr–Jun 2024
- AWS Academy — AI & ML Virtual Internship, via ACTE — Jan–Mar 2024
- Oracle — Machine Learning on Oracle Cloud — Jan 2025
- CITI Program — Human Research Curriculum (Data & Specimens Only Research), incl. 2 refresher courses affiliated with MIT — valid through Feb 2028

ACADEMIC RESEARCH

EdgeSign: An IoT-Enabled Edge–Cloud Hybrid System for Real-Time Sign Language Translation

Co-authored with Surya Sathyanarayanan and Dr. R Babu, Dept. of Computational Intelligence, SRM Institute of Science and Technology | Unpublished — internal/coursework paper, not submitted to a journal or conference

- Co-developed an assistive communication system using a Raspberry Pi 4 and Camera Module 3 to capture continuous hand-gesture movements in real time.
- Used MediaPipe Hands to extract 21 hand landmarks per frame for gesture classification, and designed an AI-driven sentence-generation layer that converts recognized gesture sequences into grammatically coherent sentences rather than isolated words.
- Implemented on-device text-to-speech synthesis on the Raspberry Pi, producing a low-latency, portable, standalone assistive communication device suited to educational, clinical, and social settings.

Medical Insurance AI Assistant

Course: Intelligent Systems for IoT Applications, SRM Institute of Science and Technology | Unpublished — internal/coursework project — Apr 2025

- Explored AI and OCR techniques to automate medical document analysis for insurance claim processing.